

Result 2.

Hydrodynamic entropy of GraphCast: spherical-harmonic power of the latent field

1 What is being measured

Result 1 assigns an energy to each mesh length by applying the mesh-edge encoder to the edges of that length. This result assigns the energy differently. GraphCast Small is run for one forecast step, and the latent vector on each mesh node is kept after the encoder and after every processor step. That field is expanded in spherical harmonics. $E(\ell)$ is the power at degree ℓ .

The same entropy is then computed from those powers, grouped into the six mesh levels of this checkpoint:

$$p_i = \frac{E_i}{\sum_j E_j}, \quad H = - \sum_i p_i \ln p_i, \quad H_n = \frac{H}{\ln N}.$$

H is in nats. Six bands give $\ln 6 = 1.792$. The wavenumber of band m is $k = 1/L(m)$, with the same lengths as in Result 1,

$$L(m) = \frac{\pi R}{2 \cdot 2^m}, \quad R = 6371 \text{ km},$$

from 10008 km at M_0 to 313 km at M_5 . Shorter lengths are absent from this checkpoint and are left out.

2 Procedure

The run uses the published GraphCast Small checkpoint: 1.0°, 13 pressure levels, mesh size 5, latent size 512, 16 processor steps. Nothing is retrained. The input is the published ERA5 state for 2022-01-01 at 1.0° with 13 pressure levels, normalized with the checkpoint’s own per-level mean and standard deviation. The target is the atmosphere six hours later.

After the grid-to-mesh encoder, and after each of the 16 processor steps, the latent on the mesh nodes is stored. For this mesh that is a matrix of shape 10242×512 . The decoded 2 m temperature is kept only as a check that the forecast step ran. Its global mean on this date is 276.9 K.

Each latent channel is interpolated from the mesh nodes onto the Gauss–Legendre grid whose degree matches the node Nyquist,

$$\ell_{\max} = \lfloor \sqrt{10242} \rfloor - 1 = 100,$$

by inverse-distance weighting from the eight nearest mesh vertices. The channel is expanded in orthonormal spherical harmonics, and the power at degree ℓ is the sum of squared coefficients over order and over the 512 channels. The degree $\ell = 0$ term is the spatial mean of the latent and is left out of H .

Degrees are grouped into octave bands aligned with the mesh. A spherical harmonic of degree ℓ has wavelength $2\pi R/\ell$. Setting that equal to $L(m)$ gives the characteristic degree $\ell(m) = 2^{m+2}$. Band M_m runs from $\ell(m-1) + 1$ through $\ell(m)$, with the first band starting at $\ell = 1$:

	M_0	M_1	M_2	M_3	M_4	M_5
degrees	1–4	5–8	9–16	17–32	33–64	65–100

The characteristic degree of M_5 is 128, which lies above the node Nyquist, so that band stops at $\ell = 100$. The energy of a band is the sum of $E(\ell)$ inside it. The slope quoted as “band” is the fit of that band energy against $k = 1/L(m)$. The slope quoted as “degree” is the fit of $E(\ell)$ itself for $\ell \geq 2$.

3 Results

H rises at every processor step, and the share of band energy on M_0 falls at every step.

Step	H (nats)	H_n	band slope	share on M_0
Encoder	1.218	0.680	−1.525	55.0%
1	1.253	0.699	−1.462	52.9%
2	1.269	0.708	−1.423	52.1%
3	1.283	0.716	−1.380	51.6%
4	1.295	0.723	−1.355	51.1%
5	1.303	0.727	−1.333	50.8%
6	1.312	0.733	−1.316	50.3%
7	1.322	0.738	−1.297	49.7%
8	1.331	0.743	−1.296	48.9%
9	1.338	0.747	−1.286	48.5%
10	1.347	0.752	−1.262	47.9%
11	1.354	0.756	−1.264	47.3%
12	1.360	0.759	−1.249	47.0%
13	1.381	0.771	−1.232	45.2%
14	1.394	0.778	−1.203	44.5%
15	1.442	0.805	−1.146	39.7%
16	1.519	0.848	−1.015	28.5%

The coefficient of determination of the band fit is 0.90 at the encoder, 0.87 at step 8, and 0.76 at step 16. By the last step the six probabilities are 0.285, 0.251, 0.261, 0.146, 0.051, 0.006. From M_0 to M_2 the band energies are then close, and the decay sits in M_3 – M_5 . Inside the bands the per-degree spectrum is steeper: $E(\ell) \propto \ell^{-2.35}$ at step 16, with coefficient of determination 0.91. At the encoder that per-degree slope is −2.78.

4 Comparison with the local baseline

The notebooks in `Main codes` are the baseline, on the same GraphCast Small checkpoint and the same six lengths M_0 – M_5 . They use only `linear_0` of the mesh-edge encoder, the 4×512 matrix. The input of an edge is $(\Delta x, \Delta y, \Delta z, \|\Delta\|)$, and the energy of a level is the mean of $\|W_0 \mathbf{x} + \mathbf{b}_0\|^2$

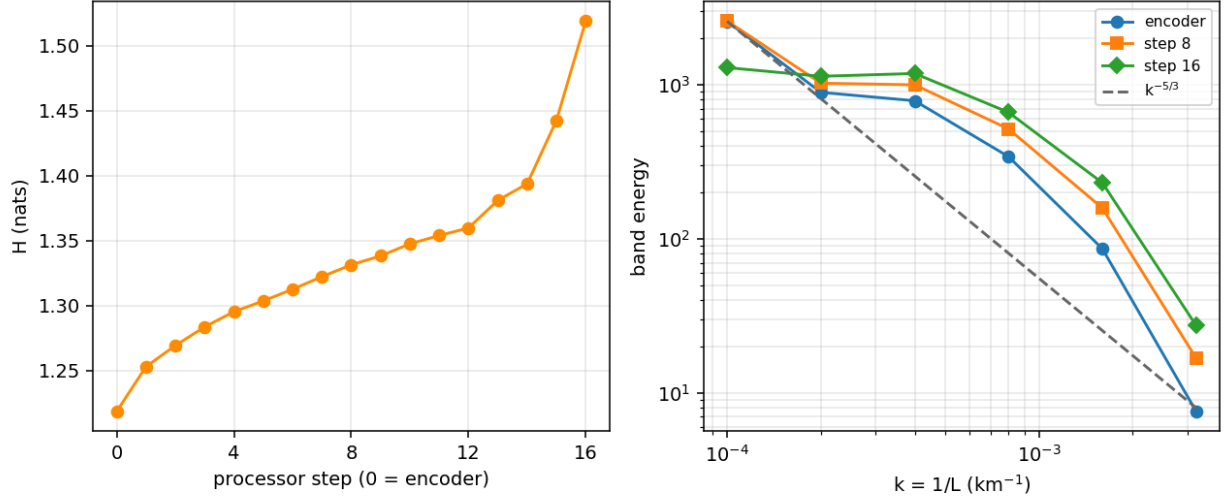


Figure 1: Left: entropy of the six band energies after the encoder (step 0) and after each processor step. Right: band energy against $k = 1/L$. The dashed line is $k^{-5/3}$, drawn through the encoder’s M_0 band.

over the edges of that level. No ERA5 field is loaded, and the second linear map, the layer norm, the 16 processor steps, and the decoder are not evaluated. Those notebooks report the entropy in bits. The row below converts that number to nats: 1.205 bits is 0.835 nats. $H_n = 0.466$ is unchanged by the change of base.

	Local notebooks	This calculation
Checkpoint	GraphCast Small, published weights	The same checkpoint
Input	Four geometric numbers on each mesh edge	Normalized ERA5 state for 2022-01-01
Depth	First matrix of the mesh-edge encoder	Encoder output and all 16 processor steps
E	Mean squared output of the first matrix	Band-summed spherical-harmonic power
Lengths	M_0 – M_5	The same six lengths

	H (nats)	H_n	slope	share on M_0
Local <code>linear_0</code>	0.835	0.466	−1.866	70.5%
Latent field, encoder	1.218	0.680	−1.525	55.0%
Latent field, step 16	1.519	0.848	−1.015	28.5%

The local slope is the fit of mean $\|W_0\mathbf{x} + \mathbf{b}_0\|^2$ against $k = 1/L$, with coefficient of determination 0.9995. The other two slopes are the band fits already quoted above. On the local baseline the six probabilities are 0.705, 0.217, 0.058, 0.015, 0.004, 0.001. After processor step 16 they are 0.285, 0.251, 0.261, 0.146, 0.051, 0.006.

The baseline value 0.835 nats is fixed by the mesh. A long edge has a much larger input vector than a short one, and $\|\mathbf{x}\|^2$ scales as length squared. Length halves at each refinement, so the energy falls by about a factor of four and the slope sits near -2 . About 70% of that energy lies on M_0 . The same entropy, to three decimals, appears for the operational and full checkpoints in those notebooks, which is what a shared geometric input produces.

The latent-field entropy uses the same formula on a different energy. At the encoder, $H = 1.218$ nats and $H_n = 0.680$, because M_0 holds 55% rather than 70% and the middle bands carry a real share. The vector being measured is the latent weather state on the mesh nodes. Edge length does not set its size. The further rise from 1.218 to 1.519 nats is the processor. Each of the 16 steps mixes that latent across edges of every length, and by the last step M_0 , M_1 , and M_2 hold comparable power. The local calculation has no depth, so it has no such trajectory.